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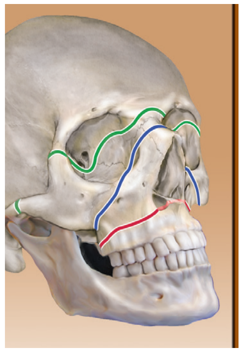

The Athlete With Diabetes

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DEFINITION

Diabetes mellitus is a disorder of glucose metabolism that affects millions of athletes of all types around the world. It is caused by either an absolute (type 1 diabetes) or relative (type 2 diabetes) deficiency in insulin, the principal hormone that regulates carbohydrate and fat metabolism. Diabetes mellitus is one of the most common chronic medical illnesses to be encountered by sports medicine specialists, coaches, and athletic trainers who work with adolescent and young adult athletes. Two other classifications of diabetes, *gestational* and *overt,* are both associated with pregnancy and are also caused by a relative insulin deficiency. In the former, glucose metabolism is normal at the time of conception and early in pregnancy. However, as the pregnancy progresses, glucose intolerance develops as a result of placental hormones that induce insulin resistance in the mother. The latter category is a recent classification which reflects that in women at high risk for diabetes, glucose intolerance or frank diabetes likely existed at the time of conception or early in pregnancy but had been undetected or undiagnosed.[1] Although women with gestational or overt diabetes may represent only a small fraction of the athletes competing with underlying diabetes, good control of blood glucose, in part achieved by regular physical activity, is especially important to protect the health of the mother and the fetus.

Exercise and athletic activity, together with medication and diet, have long been cornerstones in the treatment of diabetes and the subsequent prevention of long-term complications that affect the eyes, kidneys, cardiovascular, and nervous systems. However, the alterations in metabolism that occur during athletic activity—specifically those related to the control of blood glucose—can often present a significant challenge for the athlete with diabetes, along with his or her coaches and care providers. Athletic activity, if not properly managed, can increase the risk for the short-term complications of hypoglycemia or significant hyperglycemia, especially in athletes with type 1 diabetes. Both of these metabolic disturbances can have a negative impact on athletic performance. Athletes with diabetes must commit to understanding their illness, and they must expend considerable effort to record and analyze the impact of different activities on their blood glucose levels. In addition, all persons involved in the care of an athlete with diabetes need to have a basic understanding of the metabolic changes that accompany the disorder, how they are influenced by exercise, and methods of testing and treating diabetes so they can promote both good health and ideal performance.

EPIDEMIOLOGY

The prevalence of diabetes in American adults has increased markedly in recent decades, from 0.2% in the 1980s, 3.5% in 1990, 7.9% in 2008, and 8.3% in 2012.[2] Recent data published by the Centers for Disease Control and Prevention also show alarming incidence in children and young adults in the United States.[3] As of 2012, 29.1 million children and adults, or 9.3% of the total population, had either type 1 or type 2 diabetes. Eight million of these persons were living with undiagnosed diabetes. Furthermore, 37% of US citizens aged 20 or older have “prediabetes,” determined by a fasting glucose or hemoglobin A1c that is elevated but not yet diagnostic for diabetes. Many of these 86 million people are at risk for developing type 2 diabetes; importantly, lifestyle modifications, and in particular exercise, can substantially lower this risk.

The prevalence of diabetes in children and young adults is also striking: nearly 1 in every 400 has diabetes, representing 0.25% of this population.[3] In 2008–09, more than 18,000 people aged 20 or younger in the United States were diagnosed with type 1 diabetes and more than 5000 aged 20 or younger were diagnosed with type 2 diabetes. Children of African-American, Hispanic/Latino, Native American, Asian, and Pacific Islander descent are more likely to be diagnosed with type 2 rather than type 1 diabetes. Type 1 diabetes is diagnosed at the highest rate in non-Hispanic white children. Because of the substantial increase in the number of children, adolescents, and young adults with diabetes, it is quite likely that persons involved in the management of sports will encounter, with increasing frequency, athletes who have diabetes.

DIAGNOSIS

Most athletes with diabetes are already aware of their disease when they enroll in athletic activity. However, because the incidence in children, adolescents, and young adults is considerable, it is important that coaches and athletic trainers be aware of the symptoms of hyperglycemia. When an athlete experiences a significant (>180 mg/dL) and persistent elevation of blood glucose levels, the resultant osmotic diuresis leads to dehydration and a hyperosmolar state. Symptoms of hyperglycemia include blurry

Abstract

Diabetes mellitus is among the most commonly diagnosed medical conditions, and athletes with diabetes face unique challenges in striving for peak physical performance. Accordingly, medical professionals, coaches, and athletic trainers should be prepared to encounter this disorder in their athletes of all ages. Type 1 diabetes mellitus, caused by complete insulin deficiency, most often is diagnosed in children and young adults. As overweight and obesity rates rise, type 2 diabetes, caused by a relative insulin deficiency, is being diagnosed more frequently in both youth and adults. Exercise is a cornerstone of glycemic management in diabetes; however, glucose abnormalities may be exacerbated by exercise and can impact athletic potential. For athletes with diabetes, hyperglycemia and hypoglycemia must be recognized and treated appropriately to optimize athlete safety and performance. The management of diabetes includes dietary modification, physical activity, and often use of oral and/or injectable pharmacologic agents. Insulin is absolutely required for athletes with type 1 diabetes and is also used in many athletes with type 2 diabetes. Insulin can be administered by subcutaneous injection or via an insulin pump. Before, during, and after exercise, glucose monitoring should be performed regularly by finger-stick or optimally by a continuous glucose monitoring device. Athletes may need to modify carbohydrate intake and medical therapy use in anticipation of and in response to physical activity. On the horizon, emerging closed-loop artificial pancreas systems are expected to become the standard of care for athletes who depend on insulin.

Keywords

diabetes mellitus
insulin
glucose
hyperglycemia
hypoglycemia
exercise

vision, polyuria, nocturia, and polydipsia. Coaches and athletic trainers associated with the athlete are more likely to first notice subtle signs of hyperglycemia, such as easy fatigability or malaise. A thorough evaluation and the initiation of treatment at that time may help prevent the more serious complications of hyperosmolar coma or diabetic ketoacidosis. Although it would be difficult for an athlete to continue to practice or play regularly in the setting of serious metabolic disturbance, these critical states will eventually develop in the absence of treatment.

In the presence of any of the symptoms of hyperglycemia, a random glucose level of greater than 200 mg/dL is diagnostic of diabetes mellitus.[4] Both a fasting plasma glucose greater than 126 mg/dL (repeated once for confirmation) or a 2-hour glucose value of greater than 200 mg/dL after a 75-g oral glucose load are also diagnostic. However, a more convenient method is the measurement of a hemoglobin A1c level, a blood test that calculates the average blood glucose level over the previous 2 to 3 months. Values of ≥6.5% convey the diagnosis of diabetes,[5] although one recent report suggested that hemoglobin A1c levels may differ depending on race and should be interpreted with caution in certain groups.[6] These tests establish only the diagnosis of diabetes and do not distinguish between type 1 and type 2. The distinction is generally determined by clinical characteristics or response to medications. Patients with type 1 diabetes generally are lean and active and usually have few or no other family members with the disease. They may have or be at risk for other autoimmune disorders. Patients with type 2 diabetes are generally obese and sedentary, have a strong positive family history of diabetes, and are often members of a high-risk ethnic group, such as African American, Asian, Asian American, Latino, or Pacific Islander. In past decades, patients with type 1 diabetes were generally younger at the time of diagnosis and those with type 2 diabetes were generally middle-aged or older, but this distinction has been blurred considerably in recent years. It is now recognized that patients with type 1 diabetes can be diagnosed as older adults. In addition, in some regions of the United States, a child with newly diagnosed diabetes is more likely to have type 2 than type 1 diabetes.

Monitoring After Diagnosis

Monitoring technology allows athletes with diabetes to know the value of their blood glucose levels in real time. The most widely used method employs a handheld device (a glucose meter) that measures capillary glucose, which is obtained by lancing the skin. The meters are thought to be highly accurate except when measuring values in the extreme ranges of hypoglycemia or hyperglycemia. The main disadvantages for athletes is that a single measurement does not show the rate or direction of change of glucose over time and that most athletic activity must be stopped to perform a measurement. Over the past decade, a second type of technology, continuous glucose monitoring (CGM), has been extensively developed. An enzyme-coated filament is placed under the skin, where it reacts with tissue glucose levels. The results are wirelessly communicated every few minutes to either a separate handheld device, smartphone, or an insulin pump. This method is clearly superior to a traditional glucose meter because the athlete does not have to stop athletic activity for a measurement to be obtained. Moreover, the rate of change in glucose levels, including risk for impending hypoglycemia can be calculated and portrayed on the receiving device's screen. Several organizations, including the Endocrine Society, have endorsed CGM as the "gold standard" of monitoring for people with type 1 diabetes.[7] With advances in technology, CGM data now can inform automated pump processes. In 2016 the US Food and Drug Administration (FDA) approved the Medtronic MiniMed 530G, a pump with a low-glucose threshold suspend feature, meaning that when glucose is sensed below a set threshold, insulin infusion halts. In early 2017 the first CGM to communicate with an insulin pump to control the rate of infusion of insulin became commercially available (see *Treatment*, later).

In addition to its use as a diagnostic test, hemoglobin A1c levels can be followed to determine the degree of maintenance of blood glucose levels within a target range. In general, a hemoglobin A1c level of 6.5% or less is considered to represent good control, with control becoming increasingly poor as the level increases. When good control is maintained, athletes should be able to exercise or compete safely without concern for sudden deterioration of glucose control or ketone production, which is especially relevant for athletes with type 1 diabetes. Each athlete should consult with his or her health care provider to determine if exercising or competing is safe when glucose values are elevated.

Urine ketone testing is recommended when glucose values are consistently elevated (>250 mg/dL), or before an athletic event of significant intensity or endurance when the production of ketones may be expected. It is carried out with a reactive strip dipped into a fresh urine sample and uses a reaction of alkali and nitroprusside, which provides a semiquantitative measurement of the ketone acetoacetate.[8] Any athlete with ketones present in urine prior to training or intense competition should suspend athletic activity until the ketonuria is resolved.

TREATMENT

An in-depth discussion of all the treatment modalities used for persons with diabetes mellitus is beyond the scope of this chapter; however, each person involved in the care or performance of an athlete with diabetes should be familiar with the basic families of medications and the expected actions and potential problems with their use during athletic competition.

Medical Nutrition Therapy

Part of the triad of treatment for patients with type 1 and type 2 diabetes involves careful attention to diet and regular exercise. Patients with diabetes are at risk for cardiovascular disease, and most patients with type 2 diabetes are overweight or obese. Therefore medical nutrition therapy is recommended not only to help normalize blood glucose levels but also to help achieve targets for blood pressure and lipids.[9] Patients with type 1 diabetes generally focus on balancing the amount of carbohydrate in meals and snacks with administered insulin, but persons with type 2 diabetes are often prescribed a lower-calorie diet to enhance weight loss. Ideally, athletes with diabetes will consult with a registered dietician to discuss individualized dietary plans accounting for their goals and personal food preferences. Specific recommendations for carbohydrate consumption include choosing from a wide variety of fruits, vegetables, legumes, and whole grains and avoiding excess intake of carbohydrate sources that contain added sugar, fat, or sodium. Fiber intake from food sources is recommended at 25 g/day for women and 38 g/day for men. Intake of sugar-sweetened beverages should be limited; with the commercial availability of low calorie or zero calorie electrolyte replacement beverages, this is now easier than it had been during athletic competition. Protein and total fat intake goals should be individualized because there is no clear evidence supporting specific targets for these macronutrients. Saturated fats should comprise less than 10% of total calories, cholesterol intake should be less than 300 mg/day, and trans fat intake should be avoided. For people with diabetes, as in the general population, there is no evidence supporting use of vitamin or mineral supplements in the absence of diagnosed deficiencies.

Oral Hypoglycemic Agents

Several types of oral hypoglycemic agents (OHAs) are available in the United States. Most are not useful in the treatment of type 1 diabetes because some endogenous insulin secretion is necessary for them to be clinically effective in this population.

- *Sulfonylureas* are insulin secretagogues. They increase endogenous insulin levels in the absence of food intake and can cause hypoglycemia, especially if taken prior to exercise. Glucose levels should be carefully monitored during training and competition so the athlete can decide whether reducing or withholding their use is necessary to prevent hypoglycemia.
- *Metformin* is a member of the biguanide family and is an insulin sensitizer. Although it does not lead to an increase in endogenous insulin levels, like sulfonylurea agents, its effect should be carefully monitored in athletes with type 2 diabetes to determine if it should be reduced or withheld during training and competition.
- *Alpha-glucosidase inhibitors* alter the absorption of starch from the gut. This effect leads to a slower and smaller increase in postmeal glucose excursion. Patients have had difficulty tolerating these agents because of the adverse effects of flatulence and diarrhea.
- *Thiazolidinediones* are peroxisome proliferator-activated γ-receptor agonists that enhance insulin sensitivity. Because of a number of adverse effects, their use has been restricted in the United States. The common adverse effects of weight gain and edema make their use particularly unattractive in athletes with diabetes.
- *Meglitinides* lead to an increase in insulin synthesis within pancreatic cells. Insulin is then released when it is triggered naturally by food intake. Although the theoretic risk for hypoglycemia is less than that seen with sulfonylureas, the same care should be given to individualize their use in athletes with type 2 diabetes after careful testing has been performed during training.
- *Dipeptidyl peptase-IV (DPP-4) inhibitors* are oral agents that utilize the incretin pathway (see later) to lower blood glucose levels after meals by indirectly increasing insulin secretion and suppressing glucagon release. DPP-4 agents are not believed to cause hypoglycemia, but their data regarding their use in athletes are limited.
- *Sodium glucose cotransporter 2 (SLGT2) inhibitors:* The SGLT receptors (1 and 2) promote reabsorption of glucose through the proximal tubules of the kidney.[10] When this receptor is blocked, glucose is excreted in the urine, rather than reabsorbed into the bloodstream, resulting in lower serum glucose levels. Genital and urinary tract infections are the main side effects experienced with the three available agents (canagliflozin, dapagliflozin, and empagliflozin). However, there are no published reports of the use of these agents in the setting of intense exercise. Because other effects (positive and negative) also include osmotic diuresis, natriuresis, and a lowering of blood pressure, it is the authors' opinion that these agents should be used with caution, if at all, until data support their safe use in athletes.

Ideally, as patients with type 2 diabetes lose weight, become better conditioned with exercise, and experience an increase in insulin sensitivity, they can reduce or eliminate the need for OHAs from their regimen.

Insulin and Insulin Analogs

Many patients with type 2 diabetes use insulin in addition to or in the place of OHAs to control their blood glucose levels, but patients with type 1 diabetes are dependent on exogenous insulin to live. Several types of recombinant human insulin and insulin analogs are available in the United States. Most regimens use a combination of long-acting (basal) and short-acting (bolus) injected insulin to approximate the secretory patterns found in normal islet call physiology. Subcutaneous continuous insulin infusion (SCII or insulin pump therapy) has the capacity to deliver insulin in much smaller increments in either a basal (continuous) pattern or as a bolus designed to cover mealtime excursions in blood glucose. Pumps infuse only short-acting insulin analogs, which might be an issue for athletes who want or need to suspend pump therapy for an extended period. Active efforts are underway to develop and commercialize several artificial pancreas systems, using "closed-loop" technology to sense blood glucose levels and respond with appropriate insulin doses without need for human intervention. The FDA-approved low-glucose threshold suspend system was described previously in the section on monitoring. In addition, the first closed-loop system to receive FDA approval was the Medtronic MiniMed 670G System in September 2016.[11]

The pharmacokinetics of the various insulins and their durations of action can be found in Table 18.1. Because many patients with type 2 diabetes have insulin resistance that requires large (>200 units) daily doses of insulin, regular insulin can also be prescribed in 5× and 2× concentrations to reduce the volume of insulin injected.

It is important for athletes, coaches, and trainers to be aware that insulin, as an anabolic agent, is included on the prohibited substances list of the World Anti-Doping Agency (WADA). For elite athletes requiring insulin for diabetes management, therapeutic use exemption (TUE) must be obtained. Information related to this process can be found at wada-ama.org.

TABLE 18.1 Pharmacokinetics of Insulin Preparations

Insulin Preparations	Onset	Peak	Duration
Aspart (Novolog→)	5–15 min	45–90 min	3–4 h
Glulisine (Apidra→)	5–15 min	45–90 min	3–4 h
Lispro (Humalog→)	5–15 min	45–90 min	3–4 h
Regular[a]	30–60 min	2–4 h	5–6 h
Neutral protamine Hagedorn	1–3 h	4–6 h	8–12 h
Detemir (Levemir→)	1 h	6–8 h	12–24 h
Glargine (Lantus, Basaglar→)	1 h	—	24 h

[a]Regular insulin is also available in 5× (U-500) and 2× (U-200) concentrations.

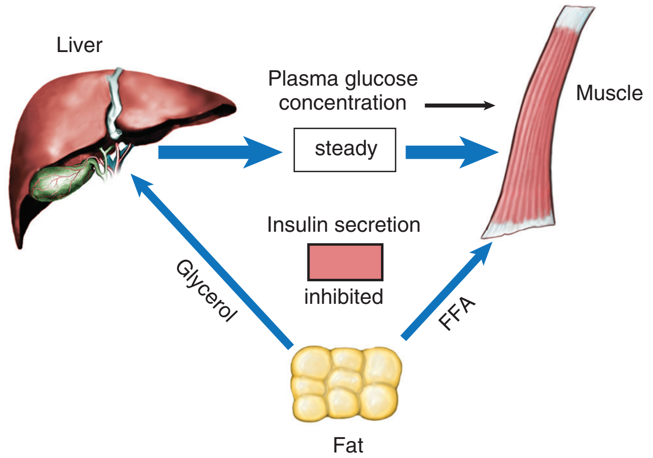

Fig. 18.1 The response to exercise in healthy persons and in insulin-dependent diabetic patients. When plasma insulin is normal or slightly diminished, hepatic glucose production increases markedly, as does skeletal muscle usage of glucose, whereas blood glucose remains unchanged. *FFA,* Free fatty acids. (From Ekoe JM. Overview of diabetes mellitus and exercise. *Med Sci Sports Exerc.* 1989;21:353–368. Copyright The American College of Sports Medicine.)

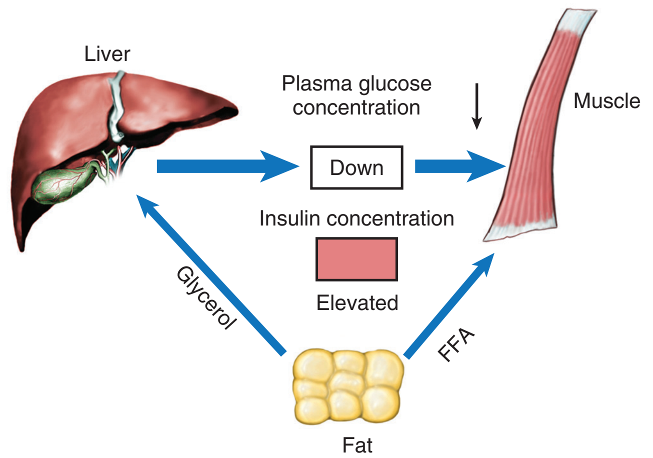

Fig. 18.2 The response to exercise in hyperinsulinemic insulin-dependent diabetic patients. When plasma insulin is increased, skeletal muscle use of glucose during exercise increases markedly, but the increase in hepatic glucose production is smaller than normal: blood glucose levels decrease. *FFA,* Free fatty acids. (From Ekoe JM. Overview of diabetes mellitus and exercise. *Med Sci Sports Exerc.* 1989;21:353–368. Copyright The American College of Sports Medicine.)

Incretin and Amylin Analog Therapy

Incretins are peptides found in the gastrointestinal tract that act to lower glucose by pathways that are independent of the insulin receptor. The two currently recognized forms are glucagon-like peptide-1 (GLP-1) and gastric inhibitory peptide. GLP-1 agonists are injectable agents that lower blood glucose by delaying gastric emptying, inhibiting glucagon secretion, and increasing insulin secretion after a meal. DPP-4 is an endogenous enzyme that inactivates the incretins. Adverse effects of GLP-1 agonists include hypoglycemia, and therefore they should be used with caution before or during exercise. Amylin is a peptide that is normally cosecreted with insulin from the pancreas and which usually antagonizes its action. The amylin analog pramlintide is available as an injectable agent and is approved by the FDA for use in type 1 and type 2 diabetes. Its method of action is very similar to that of incretin therapy, and the risk of hypoglycemia with exercise is also a concern.

PHYSIOLOGIC CHANGES OF EXERCISE IN HEALTHY ATHLETES AND IN ATHLETES WITH DIABETES

A comprehensive review of the very complex changes that occur during exercise in healthy persons can be found elsewhere.[12] However, because critical changes occur in glucose metabolism during exercise and recovery, it is important that persons who are managing athletes with diabetes have a basic understanding of what occurs during both aerobic and anaerobic metabolism.

Glucose Regulation During Exercise in Athletes With Type 1 Diabetes Mellitus

Soon after the discovery of insulin and its use in patients with type 1 diabetes, it was observed that physical activity could reduce insulin requirements. In addition, it was recognized that the decrease in blood glucose after an insulin injection was magnified by subsequent exercise.[13] Exercise increases blood flow to muscles and skin, leading to an increased rate of insulin absorption in the athlete with type 1 diabetes (Figs. 18.1 and 18.2).[14] This effect is most pronounced when insulin is administered less than 60 minutes before exercise. Because insulin is given exogenously in persons with type 1 diabetes, the body cannot decrease its release of insulin, and increased serum insulin levels inhibit hepatic glucose production and peripheral lipolysis. At the same time, continued insulin-independent glucose uptake by exercising muscles depletes energy stores. In type 1 diabetes, glucagon secretion in response to hypoglycemia is usually lost approximately 5 years after diagnosis; the epinephrine response to hypoglycemia is also attenuated in these persons.[15] Deficiencies of these counterregulatory hormones further limit fuel availability during exertion (Table 18.2). The balance between energy supply and demand is often disrupted in the diabetic athlete by an excessive or inadequate insulin effect.

TABLE 18.2 Actions of Major Counterregulatory Hormones

Hormone	Mechanism of Hyperglycemic Effect
Glucagon[a]	Activates hepatic glycogenolysis and gluconeogenesis
Epinephrine[a]	Stimulates hepatic glucose production, limits peripheral glucose use, and suppresses insulin secretion
Growth hormone	After initial glucose-lowering effect, limits glucose transport into cells, mobilizes fat, and provides gluconeogenic substrate (glycerol)
Cortisol	Initially inhibits glucose use; with time, mobilizes substrate (amino acids and glycerol) for gluconeogenesis

[a]Hormones important in recovery from acute hypoglycemia.

Pathophysiologic Responses to Exercise in Persons With Diabetes Mellitus

Data are inconclusive with regard to whether athletes with diabetes mellitus have inherently impaired exercise performance. It is possible that the associated epiphenomena of diabetes, such as acute hypoglycemia or chronic hyperglycemia, diminish the capacity of the athlete to perform at his or her highest capacity. In the setting of obesity or chronic illness, exercise performance is undeniably diminished, but in fit athletes with diabetes, particularly lean athletes with type 1 diabetes, this question remains unanswered. Therefore it is worth exploring the available literature so that both care and expectations can be managed accordingly.

In a study that compared adolescent girls who had either type 1 or type 2 diabetes with either obese or normal-weight girls without diabetes, it was found that female adolescents with type 2 diabetes have reduced aerobic capacity and reduced heart rate response to maximal exercise.[16] Moreover, investigators found that subjects with either type 1 or type 2 diabetes had a blunted stroke volume response compared with nondiabetic control subjects. However, subjects with either type 1 or type 2 diabetes had evidence of poor glucose control at baseline (hemoglobin A1c level = 8.8% and 8.2%, respectively). Moreover, no determinations of baseline activity levels were made. Therefore it is possible that the subjects with either type 1 or type 2 diabetes were more sedentary than their matched nondiabetic control subjects. This concept is supported by another study comparing aerobic capacity and pulmonary function in athletes with type 1 diabetes and both matched, nondiabetic control subjects and nonexercising, diabetic control subjects.[17] In this instance, the glycemic control of the athletes with diabetes was much better than that of the nonathletes with type 1 diabetes (mean hemoglobin A1c = 7.5% vs. 9.0%), but the average glucose level was still much higher than that of the athletes without diabetes (mean hemoglobin A1c level = 4.4%). In addition, the type and duration of programmed exercise was not described. Nevertheless, aerobic capacity (as measured by $VO_{2\ max}$) was found to be similar in athletes both with and without diabetes and was higher in both groups than in nonathletes with diabetes. Of note, both forced expiratory volume in 1 second and anaerobic threshold (as measured by nonlinear increases of pulmonary ventilation and peak CO_2 production compared with O_2 consumption) were lower in athletes with diabetes compared with nondiabetic athletes. Because aerobic capacity was normal, the authors speculated that other factors besides ventilation were adversely lowering the anaerobic threshold. They hypothesized that elastin and collagen abnormalities may negatively affect bronchial adaptation to air flow. With average glucose values significantly higher in the diabetic versus nondiabetic athlete subjects in this study (an average difference of greater than 80 mg/dL according to hemoglobin A1c levels), it is difficult to exclude the effect of chronic hyperglycemia alone on exercise performance in study subjects. In that regard, Wheatley et al.[18] reported that the diffusing capacity for carbon monoxide and the membrane diffusing capacity were lower in athletes with uncontrolled type 1 diabetes compared with athletes with better glycemic control. However, an inverse relationship between glycemic control and pulmonary function was less clear after the investigators found that arterial oxygen saturation was lower in the subjects with well-controlled diabetes.

Other investigators have used additional technologies to observe athletes with diabetes. Peltonen et al. used near-infrared spectroscopy to study local tissue deoxygenation rates during cycling in men with type 1 diabetes (mean hemoglobin A1c 7.7 ± 0.7%) and healthy control men, who were matched for age, body measurements, and baseline reported physical activity.[19] Despite similar reported baseline physical activity, the men with type 1 diabetes were found to have lower aerobic capacity ($VO_{2\ peak}$). As a novel finding in this population, the men with diabetes had more rapid leg muscle deoxygenation at submaximal workloads. Because there was no difference in arterial oxygen saturation in the two groups of men, the authors noted that impaired alveolar gas transfer was an unlikely contributor to this observation. The investigators suggested that this finding instead could indicate inadequacy in circulatory capacity to amplify oxygen delivery to meet increasing tissue demand.

Baldi et al. attempted to directly address the issue of glycemic status on exercise response in endurance athletes with type 1 diabetes.[20] They reported no difference between aerobic capacity or cardiopulmonary exercise response between athletes with type 1 diabetes compared with subjects who did not have diabetes. However, significant differences were found between athletes with diabetes when the group was stratified into those with good glycemic control (hemoglobin A1c level <7%, mean = 6.5%) and those with poor glycemic control (hemoglobin A1c level >7%, mean = 7.8%), despite a similar time spent exercising per week. Multiple indicators of cardiopulmonary fitness were altered in the group with poorer glycemic control, including a lower resting cardiac output and higher systemic vascular resistance. In addition, the group with poor glycemic control had a 24% lower workload during peak exercise, a 10% lower $VO_{2\ max}$, and a 25% lower calculated cardiac output. All measured parameters of pulmonary function were lower in the group with poor glycemic control. When interpreting these findings, it is important to note that in most clinical settings, a hemoglobin A1c level of 7.0% to 7.8% would indicate glycemic control that was only minimally above target.[5]

Finally, Nguyen et al. studied children with and without type 1 diabetes to assess fitness levels in the setting of poor glycemic control (hemoglobin A1c ≥9.0% for the 9 months prior to the study period) versus good glycemic control (hemoglobin A1c ≤7.5% over the 9 months prior).[21] The investigators assessed the children's grip strength, short-term muscle power during cycling, aerobic capacity while cycling, and physical activity while wearing an accelerometer for 7 days. They found no significant differences between groups for grip strength or short-term muscle power but identified lower $VO_{2\ peak}$ in the children with poorly controlled diabetes as compared with the controls or children with well-controlled diabetes. The groups of children had no significant differences in physical activity by minutes of activity per day or by activity intensity level.

If poor glycemic control has a direct negative effect on exercise performance, several potential explanations exist. First, even acute elevations in blood glucose levels are known to diminish autoregulation of capillary blood flow, leading to diminished oxygen consumption.[22] In addition, chronic hyperglycemia is associated with neuropathic changes, both autonomic and peripheral. Veves et al. noted reduced aerobic capacity and diminished heart rate in subjects with type 1 diabetes and neuropathy compared with subjects without neuropathy.[23] As the distinction between exercise performance in athletes with good glycemic control compared with poor glycemic control becomes better defined, it is expected that additional mechanisms that might explain these differences will be determined.

MANAGEMENT OF ATHLETES WITH DIABETES DURING TRAINING AND COMPETITION

Despite the fact that exercise is strongly recommended as a mainstay of therapy for persons with diabetes mellitus and that chronic athletic conditioning evokes positive metabolic changes, studies have shown a lack of improvement in hemoglobin A1c levels in patients with type 1 diabetes.[24] Several possible explanations exist for this finding. First, exercise introduces variables of glucose utilization and insulin sensitivity that may cause difficulty in keeping glucose levels consistent, especially if exercise is not performed regularly. Perhaps more important is the justifiable concern of hypoglycemia both during but especially after competition. Anecdotally, concern for developing hypoglycemia leads many recreational, high school, and even collegiate athletes to induce hyperglycemia prior to training or competition either by consuming an excessive amount of simple carbohydrates or by underdosing insulin. The resultant hyperglycemia can offset any positive effects of athletic activity on long-term glycemic control. Therefore perhaps the most important advice for helping athletes manage their blood glucose levels is to encourage each athlete to meticulously record glucose levels and the conditions that are present during different types of training or competition. As an adjunct to capillary glucose values, the use of CGM may help to inform a more complete understanding of an athlete's glucose trends before, during, and following exercise; importantly, CGM also can help to reduce athletes' fear of developing exercise-induced hypoglycemia.

Altogether, glycemic data help the athlete and his or her care provider recognize patterns of glucose control and implement nutritional or pharmacologic strategies during athletic activities to prevent hypoglycemia or hyperglycemia. A prototype glucose log for this purpose is shown in Fig. 18.3. In addition, multiple apps and websites are available to help athletes with diabetes. A particularly useful site is *Excarbs*, developed by a University of Toronto team led by professor Michael Riddell. Recreational athletes can enter values for current glucose and the time and dose of last insulin bolus taken, and the app will recommend either additional carbohydrates or reduction in insulin to prevent hypoglycemia.[25]

Training Versus Competition

It has long been known that different intensities of athletic activity lead to different compensatory hormonal responses,[26] including those of the counterregulatory hormones epinephrine, cortisol, and growth hormone. Shetty et al. studied nine adolescents and young adults with type 1 diabetes at four different exercise intensities on 4 separate days.[27] At low-to-moderate exercise intensities, they observed that exogenous glucose was necessary to maintain euglycemia, whereas at high exercise intensities ($VO_{2\ peak}$ >80%) exogenous glucose requirements were absent. Furthermore, athletes at every level of competition have anecdotally reported that insulin requirements and postevent glucose excursions are generally higher on days of actual competition compared with days of training, despite the fact that the same activity is performed for the same interval of time. Little information exists in the scientific literature to explain this phenomenon. Indeed, it would be difficult to test in a clinical laboratory setting. Even if the intensity or duration of athletic activity was closely replicated, it would be challenging to control for the emotional response or other psychological elements that accompany actual competition. It is important for athletes to be aware of this phenomenon, and with regular sporting activity, to make adjustments for it as they learn their own responses.

Glycemic Log for Athletic Events

Date/Time	Event and Duration		Basal insulin adjustment	Last insulin bolus/amount	Timing/CHO (g) last meal	Timing/CHO (g) during event	Glucose			Conditions (Temp, Altitude)	Performance Outcome
	Training	Competition					Before	During	After		

Fig. 18.3 A proposed log that athletes with diabetes can use to record parameters affecting glycemic control and athletic performance. *CHO,* Carbohydrate.

Management of Type 1 Diabetes

Although the American Diabetes Association has a position statement regarding exercise and diabetes management,[28] no single guideline or source of recommendations is available that athletes with type 1 diabetes can follow for management of their disease during training or athletic competition. As one elite athlete commented, it is an "independent disease," and each athlete must know how his or her body will respond during different types of athletic conditions.[29] However, some basic treatment principles and monitoring strategies can be used.

- *Glucose target:* Although the target glucose value must be determined for each individual athlete, a general recommendation would be to attempt to keep glucose levels between 100 and 180 mg/dL during training and competition. Values less than 100 mg/dL may place the athlete at risk for hypoglycemia during activity, and values greater than 180 mg/dL are above the renal threshold for glucose. The resultant osmotic diuresis may lead to additional loss of fluids and electrolytes. Most athletes believe that they compete best when their glucose values are in the middle-to-upper portion of this target range, which allows a gradual decrease to occur without significant risk for hypoglycemia during activity. Note that athletes who use a low-glucose suspend insulin pump may be able to set their target glucoses considerably lower because their theoretic risk for hypoglycemic should be reduced or eliminated. However, data regarding this outcome are limited only to clinical trials at this time.
- *Glucose monitoring:* For athletes not utilizing CGM, capillary glucose levels should be checked before, during, and after training or competition. Until patterns are recognized, it is advisable to check at 15-minute intervals prior to exercising to determine the rate of change of blood glucose. This is especially important if the athlete has injected an insulin bolus for a meal or snack within the previous 3 hours. If possible, athletes should also check their glucose levels during exercise or competition to determine its effect. This task will clearly be less practical during actual competition, although sports such as tennis or golf that have breaks between play or team sports with scheduled breaks or the ability to substitute players make this task more feasible. Athletes competing in individual sports, especially endurance sports such as running, cycling, or swimming, may have more difficulty with this recommendation and should use the help of others to assist with monitoring at various points in the competition. Although CGM devices should provide a superior option, these devices may not perform as well during exercise as they do at rest.[30] However, a study comparing accuracy of one sensor (Dexcom G4 Platinum) during moderate versus intermittent high-intensity exercise found that, despite significant differences in mean glucose, lactate, and pH levels, the sensor was comparably accurate at both levels of activity.[31] For persons who do not want or are unable to obtain a CGM device, one can be prescribed by their health care provider for a 72-hour to 7-day diagnostic period. The device captures glucose levels every 5 minutes during this period, and a graphic printout is provided at its completion. Ideally the athlete would wear the device during both a training session and competition to maximize its utility in determining patterns of glucose levels. Finally, it is important for athletes to recognize when they are most at risk for hypoglycemia after exercise. Typically the period of highest risk occurs 6 to 15 hours following exercise, when glycogen stores are repleted.[32] By regularly checking glucose levels during this period, the athlete can know whether insulin administration should be reduced or additional carbohydrates should be consumed in the postexercise period.
- *Insulin administration:* Athletes must consider the level of insulin in their system and its duration of action when planning athletic activity. Unless a sufficient amount of carbohydrate is consumed before or during activity, it is generally ill advised to take a bolus of insulin directly prior to training or competition. The duration of action for the short-acting insulin analogs is approximately 3 hours, and the risk of hypoglycemia should be reduced if athletic activity is performed outside of this time frame. In addition, one must consider that once it is injected or infused, insulin will be bound to its receptor and biologically active for approximately 30 minutes. Therefore athletes should avoid reducing or suspending insulin pump basal rates immediately before an athletic event but rather should do so approximately 30 minutes beforehand. For new athletes or athletes beginning a new activity, basal insulin might be reduced by 50% before and during the activity until glucose patterns can be determined and a more tailored adjustment can be made. This should help prevent hypoglycemia during the event and help the athlete to avoid finishing the event with significant hyperglycemia. This objective is more easily accomplished with an insulin pump, and particularly so with sensor augmentation, meaning that when low-range glucoses are detected, the pump automatically suspends insulin infusion. Glycemic management is anticipated to be increasingly automated and safer with the emergence of closed-loop artificial pancreas systems. In research settings, artificial pancreas systems are being adapted to include mechanisms for detection of exercise. In a recent clinical trial, closed-loop systems integrated with heart rate detection appeared to improve adolescents' glycemic management during exercise, specifically by reducing the amount of exercise time with glucose levels less than 70 mg/dL.[33] As the exercise detection mechanisms and functional algorithms for artificial pancreas systems are optimized, the artificial pancreas is anticipated ultimately to become the standard of care for athletes with insulin-dependent diabetes. Particularly, it should benefit those with a propensity to become hypoglycemic or hyperglycemic during and after activity.

  Comparatively, with injections of long-acting insulins such as insulin glargine or neutral protamine Hagedorn (NPH), additional carbohydrates may need to be consumed to counterbalance the effect of exogenous insulin (see the next section). As mentioned previously, glucose levels during and after intense competition are likely to be higher than those encountered during training, but use of additional basal insulin should be considered only after the athlete is confident of his or her individual glucose patterns during these times.

- *Carbohydrate consumption:* In the authors' experience in working with athletes who have diabetes, carbohydrate consumption is the area of most personal variability between individuals. Some athletes make adjustments to their insulin and will not require any additional carbohydrates before or during competition. Others will consume a small snack immediately prior to activity: in a survey of 91 adult endurance athletes with type 1 diabetes, 37% reported consuming preexercise carbohydrate snacks most or all of the time.[34] Still others will use fast-acting glucose in the form of tablets, gels, or sports drinks at regular intervals while training or competing. In general, carbohydrates should be consumed shortly after exercise to prevent early postactivity hypoglycemia.[35] Athletes should record both the type and amount of carbohydrate consumed to determine its effect on glucose levels during and after athletic activity. Of note, one study demonstrated that more carbohydrates should be consumed on the day after exercise to prevent hypoglycemia.[36] Because this is beyond the expected period of glycogen repletion, it suggests that any increase in insulin sensitivity gained from exercise can be prolonged. Finally, the combination of CGM and an individualized algorithm for carbohydrate intake was shown to be effective in preventing hypoglycemia in children and adolescents involved in athletic activities at a summer camp.[37] It is possible that strategies for athletic activity can be optimized by utilizing several treatment modalities at once (both for monitoring and adjusting glucose).
- *Monitoring ketones:* In most situations, ketones do not need to be routinely monitored by athletes with diabetes. However, because athletic activity can worsen ketoacidosis in persons with very poor glycemic control, a general recommendation is to check urine ketones if capillary blood glucose is 250 mg/dL or greater. If ketones are present, athletic activity should be postponed until glucose levels are lowered and ketonuria is resolved.

For additional recommendations on managing athletes with type 1 diabetes from the viewpoint of the athletic trainer, see the comprehensive review by Jimenez et al.[38]

Management of Type 2 Diabetes

Although patients with type 2 diabetes are generally sedentary and obese, they are occasionally both fit and athletic. It is the relative lack of insulin that leads to hyperglycemia, and this condition can occur with any body habitus. For patients with type 2 diabetes who are embarking on a new exercise program or athletic activity, several points should be kept in mind.

- *Risk of hypoglycemia:* Although athletes with type 2 diabetes are more resistant to the effects of insulin than those with type 1 diabetes, any oral agent that increases endogenous insulin secretion or exogenous insulin injections can increase the risk for hypoglycemia if not properly managed. Therefore athletes with type 2 diabetes, especially those who are already at their ideal body weight, should use the same strategies previously listed for patients with type 1 diabetes. They should also log glucose levels and conditions and regularly analyze the recorded results to determine if and when nutritional or pharmacologic therapies need to be modified in the setting of athletic activity.
- *Screening for cardiovascular disease:* Patients with type 2 diabetes and those with type 1 diabetes who have long-standing disease (>20 years) or microvascular complications are at increased risk for undiagnosed cardiovascular disease. Before the start of any new exercise or athletic activity, a complete medical evaluation should be performed and formal graded exercise testing should be considered for persons initiating more intense activities.[39] Athletes with diabetes may take statins for primary cardiovascular risk reduction or in the setting of known cardiovascular disease. These athletes should not be discouraged from using an indicated statin or be restricted in appropriately planned exercise. For athletes engaging in training that gradually increases in intensity, muscular metabolic adaptation appears to reduce the risk of statin-induced muscle injury.[40] Statin users who engage in vigorous exercise without appropriate training may be at heightened risk of rhabdomyolysis.[41]
- *Weight loss:* For athletes who weigh more than their ideal body weight, regular exercise should promote weight loss. Gradual modifications to the athlete's pharmacologic and nutritional regimen may be needed to prevent hypoglycemia as the athlete becomes more insulin sensitive. In some cases, pharmacologic treatment may be discontinued altogether.

Additional Considerations for Elite Athletic Competition in Persons With Type 1 Diabetes

In the authors' experience, most athletes with type 1 diabetes who are engaged in recreational, high school, or collegiate sports are most concerned about becoming hypoglycemic during competition. The most common strategy is to allow or induce hyperglycemia, either by consuming extra carbohydrates with no insulin coverage or by reducing or withholding insulin. Experiencing a hypoglycemic event in the midst of competition would no doubt diminish performance. However, for the elite or professional athlete, significant or modest hyperglycemia could also be detrimental during athletic competition. Moreover, when an elite level of competition demands the most perfect of conditions, even mild hyperglycemia could negatively impact performance. There remains a notable lack of scientific investigation and clinical guidelines for glucose management during elite competition in athletes with diabetes. Individual elite athletes, often with teams of nutritionists, personal trainers, and exercise physiologists, are primarily responsible for determining the balance of insulin and nutrition that allows them to optimally train, compete, and recover from competition.

SUMMARY

Athletes with diabetes mellitus face several challenges when training for and competing in athletic activity. To avoid hypoglycemia and hyperglycemia, many variables must be considered and adjusted to keep glucose levels in an optimal range, which is especially important because exercise itself significantly affects glucose levels. All persons involved with an athlete who has diabetes, including health care providers, coaches, athletic trainers, and most importantly, the athlete, must understand the basic principles of exercise physiology and the different types of medications used in the treatment of diabetes. This understanding is especially important as more children, adolescents, and young adults are diagnosed with diabetes. Despite significant challenges encountered during training and competition, performing well in athletic activity is very possible for athletes with diabetes. It is hoped that with continued technologic advancement, both in the arena of CGM and in the delivery of insulin, including the long-anticipated development of an artificial pancreas,[42] the challenges faced by athletes with diabetes will continue to diminish.

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For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Baldi JC, Cassuto NA, Foxx-Lupo WT, et al. Glycemic status affects cardiopulmonary exercise response in athletes with type 1 diabetes. *Med Sci Sports Exerc*. 2010;42:1454–1459.

Level of Evidence:

V

Summary:

This study is an important investigation into whether the underlying condition of diabetes negatively affects athletic performance or whether the degree of glycemic control is more responsible for aberrations in exercise physiology.

Citation:

Jimenez CC, Corcoran MH, Crawley JT, et al. National Athletic Trainers' Association position statement: management of the athlete with type 1 diabetes mellitus. *J Athl Train*. 2007;42:536–545.

Level of Evidence:

V

Summary:

This article includes useful recommendations for the athletic trainer who works with patients with type 1 diabetes. In addition to providing practical suggestions regarding supplies to carry and guidelines for travel, the recommendations focus on unique management issues for athletes with type 1 diabetes such as preparticipation physicals and healing of injuries.

Citation:

Blauw H, Keith-Hynes P, Koops R, et al. A review of safety and design requirements of the artificial pancreas. *Ann Biomed Eng*. 2016;44:3158–3172.

Level of Evidence:

V

Summary:

This article provides an updated and comprehensive review of the history, ongoing investigation, and expected future use of the technologies being applied to the development of a closed-loop system in the management of blood glucose.

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